Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Jacob league	jaleague@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal will be to make a website where you can look at cars that are for sale

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are many websites that achieve the same thing like cars.com autotrader ect

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

You can see all the different options and then go into their specific pages to see the information that's specific to the vehicle laid out

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The user should be able to search what kind/specific vehicle they are looking for and be shown options relevant to their search so they can hopefully choose the correct one

2. Input: What specific information will the user need to provide to your program?
3. (e.g., a city name, a numerical budget, a list of hobbies).

They will be able to put in the make/ model or a type of vehicle and will be given options based on their search

4. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will look through the options and match the users selections with the information stored in data classes to the options of vehicles

5. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back a list of vehicles that line up with what they searched with the most relevant options being toward the top

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

What the car is, the condition, the price, where the car is

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need about the car: what color it is, what year it is, what engine it has, what trim level it is. Details about the car I can leave out: where the car is from, who the last owner was, how long it's been for sale. Details about the condition i need: how many miles, overall condition ie fair poor ect. Details about condition i dont need: are there any dings, are there any minor scratches when the last oil change was. Details I need about the price: what the price is. Details I don't need: what the seller's bottom dollar is, financing options. Details i need about where the car is: the dealerships/sellers phone number and website details about where the car is i dont need: the dealerships last sales, the owners name

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Filling the website with all the individual cars might be tedious just because the amount of information required and the amount of cars the website need to be useful

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will work on the code during the times that i usually work on my other comp sci work and if i were to use llms i would write in all of the code myself and make sure to document things so i know what certain elements do and what they are for to ensure i understand whats going on in order to fix any mistakes or issues i may encounter

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

I hope that i can make a product that seems relatively polished and is actually functional

with plenty of information and could be useful to the world if it were to be published and in doing this i will learn more about drafter and may be able to apply what i learn here so somewhere else

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

Something that is an issue when it comes to tracking study hours is just the actual amount of time that you stay focused on the thing you are working on so the actual time spent working isn't always accurate. 3 simple features a tool should have is the information a way to search and a way to view things differently

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

They came up with a way to filter searches to find exactly what you are looking for, to only show options that match exactly what you select similar to most shopping websites with search features. They also thought of a way for people to add cars for sale to the website

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I feel like adding vehicles into the website is the more complicated idea to add to the website, im not exactly sure how to add things to the website without editing the code of the website but likely with the help of llms I will be able to find a way to do it.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I feel like there hasn't been much that has changed to my idea but i will try to maybe incorporate a way to put cars up for sale along with filtering searches to get the most exactly what youre looking for

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If i get stuck on something i will likely ask an llm because of the convenience though going to office hours might be a little more helpful its a little bit more work to go all the way to in person office hours, i could also consult the discord but id rather use the other options

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will likely look over all of my code for small mistakes and trace the errors that i am getting and if i cant seem to fissure it out i might ask a classmate or go to office hours to get my issue resolved

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I didnt use an llm for this assignment i asked a friend for the questions prior

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again.

Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.